

b.

(a)

$$H_0: \sigma_M^2 = \sigma_F^2$$

$$H_1: \sigma_M^2 > \sigma_F^2$$

$$SS E_M = 97161.9194$$

$$SS E_F = \widehat{\sigma}_F^2 \cdot (4+3-4) = 5078.056$$

$$F_0 = \frac{\frac{SS E_M}{839}}{\frac{SS E_F}{419}} = 14.1$$

$$RR = \{F > F_{0.95}(573, 919) \approx 1.163\}$$

$F_0 \in RR$, we reject H_0

(b)

$$H_0: \sigma_{\text{single}}^2 = \sigma_{\text{married}}^2$$

$$H_1: \sigma_{\text{married}}^2 > \sigma_{\text{single}}^2$$

$$SS E_S = 56231.0382, SS E_M = 100703.0971$$

$$df_S = 900 - 5 = 895, df_M = 600 - 5 = 595$$

$$F_0 = \frac{\frac{100703.0971}{595}}{\frac{56231.0382}{895}} = 1.189$$

$$RR = \{F > F_{0.95}(595, 895) \approx 1.189\}$$

$F_0 \in RR$, we reject H_0

(c)

$$NR^2 \sim \chi_g^2$$

$$g = 5, NR_0^2 = 51.03$$

$$RR = \{NR > \chi_{0.95}^2(5)\}$$

$NR_0^2 \in RR$, we reject H_0

(d)

$$df = 9 + 6 + 6 = 12, NR_0^2 = 78.82$$

$$RR = \{NR > \chi_{0.95}^2(12) \approx 21.03\}$$

$NR_0^2 \in RR$, we reject H_0

(e)

Intercept, EDUC $\Rightarrow$ Wider

EXPER, METRO, FEMALE $\Rightarrow$ Narrower